Teaching a 4B Gemma to Write TPU Kernels: LoRA Fine-Tuning on Oracle-Verified Pallas Code

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***Abstract*—Pallas is the kernel language of JAX for Google TPUs. A well written Pallas kernel can outrun the XLA compiler, but the language is young and thinly documented, and general-purpose language models rarely produce a kernel that compiles. We ask whether Gemma 4 E4B, a 4-billion-parameter open model, can learn to write Pallas kernels from verified examples alone. We build a TPU oracle that accepts a solution only if it is a genuine kernel, runs at full size on a TPU v6e, matches a float32 reference on three seeds, and reaches at least 0.7× the speed of XLA. With it we assemble 2,304 training conversations whose targets were all judged correct on the chip, written by an open-weights teacher model and a template generator, and we train a rank-64 LoRA adapter. On the 58-task primary set of JAXBench, with three attempts per task, the base model passes no task with or without 25k tokens of Pallas documentation in its prompt. The adapter passes 9 (exact McNemar $p = 0.004$), which is statistically indistinguishable from the 7 passed by Gemini Flash with the same documentation ($p = 0.79$). The two models succeed in different ways: Flash writes a numerically correct kernel for 36 tasks but only 7 are fast enough, whereas the adapter is correct on 9 and all 9 run faster than XLA (1.30× to 1.39×). Eight of the nine are tiled matrix-multiply kernels, the family our training data targets at the benchmark's scale, and the skill does not yet extend to array sizes that the learned tiles do not divide.**

***Index Terms*—code generation, fine-tuning, LoRA, accelerator kernels, TPU, Pallas, JAX, benchmark evaluation**

I. Introduction

Most numerical code that runs on an accelerator is compiled automatically. In JAX, a function is traced and handed to XLA, which chooses how to schedule each operation on the device [1]. When the automatic result is not fast enough, an expert writes a kernel by hand. On Google TPUs this is done in Pallas, a JAX extension in which the programmer states how arrays are cut into blocks, which blocks are loaded into the chip's small fast memory (VMEM) at each step, and what is computed on them [2]. A good kernel keeps the matrix unit busy, never asks for more VMEM than exists, and fuses operations that XLA would run as separate passes over memory.

Writing such kernels is specialist work, and language models do it poorly. Pallas has little public code, its lowering rules for TPU are strict, and an error in a block shape is only discovered on the chip. JAXBench [3] measures this directly: it gives a model a plain JAX function and asks for an equivalent Pallas kernel. Its authors report that most failed samples are API misuse or compile errors, and they name fine-tuning on curated Pallas corpora as future work. For GPUs, the analogous problem has received a benchmark [4] and several trained models [6], [7], [10]. To our knowledge no fine-tuned Pallas model has been published, and no result on JAXBench exists for a small open model.

This paper tests whether a small model can acquire the skill through its weights. We fine-tune Gemma 4 E4B [11] with LoRA [13] and compare it with the same model given an identical prompt, so that any difference is attributable to the adapter. We deliberately give the fine-tuned model no documentation or examples in its prompt. As a reference point we also run a much larger commercial model, Gemini Flash, with documentation.

Our contributions are as follows.

- A TPU oracle with a static kernel check, full-size execution, three-seed correctness and a speed bar, used unchanged for both training-data admission and evaluation (Section III-B).
- A training set of 2,304 conversations in which every target kernel was judged correct on a TPU, built without any closed-model output, and screened against the benchmark at the level of the computation rather than the text (Sections III-C and III-D).
- A serving finding: an adapter that will be served on 4-bit weights must be trained on, and folded into, those exact quantized weights (Section III-E).
- An evaluation in which the adapter raises the base model from 0 to 9 of 58 JAXBench tasks, matching Gemini Flash with documentation on the pass metric, together with an account of what the adapter did and did not learn (Sections IV and V).

II. Related Work

Kernel benchmarks. KernelBench [4] asks models to rewrite PyTorch modules as fast CUDA kernels and scores correctness and speedup. Follow-up work on verified variants of that benchmark showed that weak tests allow models to score by exploiting the checker rather than by solving the task [5]. JAXBench [3] carries the idea to TPUs: its level 1 contains 50 workloads, 33 operator chains ported from KernelBench (k-tasks) and 17 production kernels such as attention variants and normalizations (p-tasks). Its authors report that with curated documentation and iterative refinement of up to 144 attempts per task, a frontier model solves 48 of the 50. None of these works trains a model.

Trained kernel models. KernelLLM [6] fine-tunes an 8B Llama model on PyTorch-to-Triton pairs. AutoTriton [7], TritonRL [8], Dr. Kernel [9] and Kevin [10] add reinforcement learning from execution feedback, all for GPUs. Several of

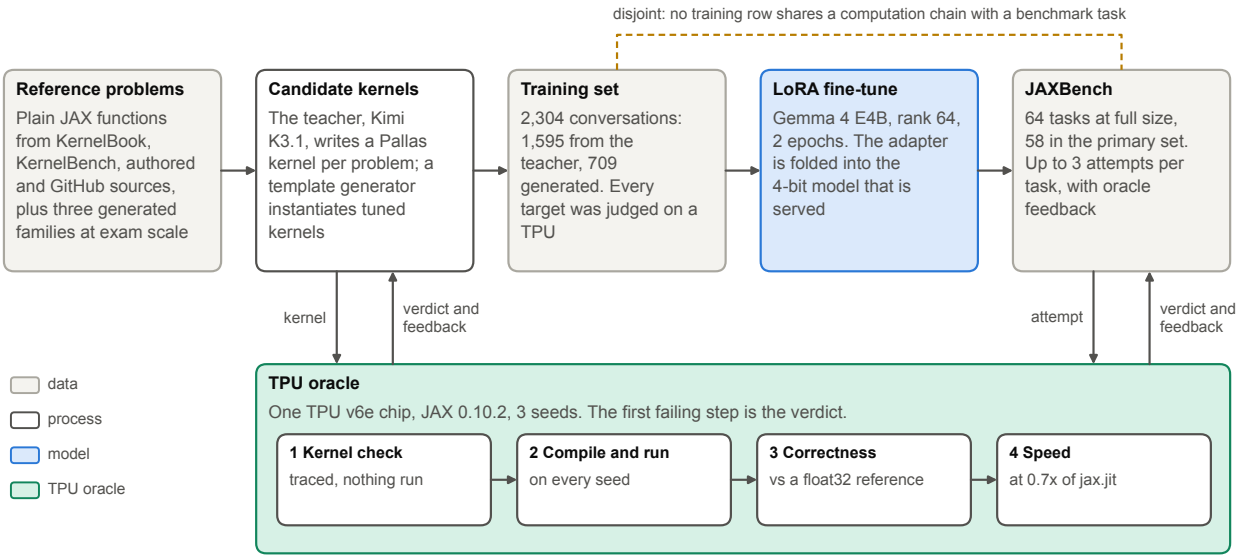


Fig. 1. Overview. Training targets come from two sources, both judged on a TPU. A LoRA adapter is trained on Gemma 4 E4B, and the same oracle scores the fine-tuned model on JAXBench.

them add guards against reward hacking; the static kernel check in our oracle plays that role for TPUs.

Parameter-efficient fine-tuning. LoRA [13] trains low-rank updates to frozen weight matrices. We use it unchanged. The only departure from common practice is where the update is applied, which Section III-E explains.

III. Method

A. Task format

A task is a plain JAX module in the JAXBench layout. A dictionary `CONFIG` holds every size, `create_inputs()` returns seeded arrays, and `workload(*inputs)` is the reference computation. A solution is a module that defines `workload` with the same arguments and performs its main computation inside `pl.pallas_call`.

B. The TPU oracle

Every solution in this paper, whether a candidate training target or a benchmark attempt, is judged by one program (Fig. 2). It applies four tests in order and reports the first failure.

Kernel check. The solution is traced without being run. It must contain a `pallas_call`, the output must depend on that call, and outside the call only layout and dtype primitives and small integer metadata are allowed. A solution that computes the answer in ordinary JAX and wraps a token kernel around it is rejected as `not_a_kernel`. This test is static, so it gives the same verdict on a CPU.

Execution. The solution is compiled and run on a TPU v6e at the task's full size. Failures are reported as `compile`, `vmem` (the blocks do not fit on-chip memory), `mosaic` (the TPU backend rejects the kernel), `timeout` or `runtime`, with the chip's error text.

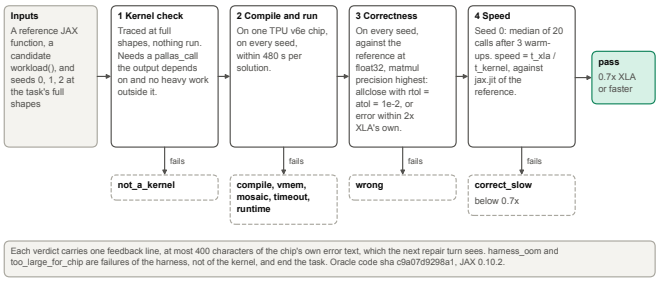


Fig. 2. The oracle's four tests and the verdict each failure produces.

Correctness. The output is compared on three input seeds with a float32 reference computed at the highest matmul precision. It must agree to absolute and relative tolerance 10^{-2}, or have at most twice the error that XLA's own compiled reference has. Otherwise the verdict is `wrong`.

Speed. Let t_{xla} and t_{kernel} be the medians of 20 timed calls of the jitted reference and of the kernel. A correct kernel with $t_{\text{xla}}/t_{\text{kernel}} \geq 0.7$ receives `pass`; a correct but slower one receives `correct_slow`. The bar of 0.7 is a policy choice: it demands a kernel that is competitive with the compiler without requiring that a first attempt beat it.

All results use the same oracle files (hash `c9a07d9298a1`), JAX 0.10.2 and a single v6e chip. Some faulty kernels halt the TPU core, after which every later item in the same process fails. Our harness re-runs those later items in a fresh process and never records the collateral failures as verdicts.

C. Source problems and contamination control

Training problems come from four places: KernelBench levels 1 and 2 and the KernelBook collection, both ported from PyTorch to JAX and checked numerically against the original module; operators written for this project, including attention variants, mixture-of-experts routing, quantized matmuls and state-space scans; and JAX references for publicly available Pallas kernels. Each problem must pass CPU checks at full size before use: it loads, fits device memory, is deterministic for a seed, varies across seeds and produces finite output.

JAXBench is used only as an exam. Because JAXBench's k-tasks were themselves ported from KernelBench, a text filter is not enough to keep exam content out of training, and we exclude a training problem when it is the same kernel as any of the 73 JAXBench files under three tests. The first removes copies: KernelBench problems that JAXBench ported, and near-duplicate text measured on two token streams. The second compares computations. We trace each `workload` to JAX's intermediate representation, reduce it to its chain of primitives, set aside linear arithmetic and layout operations, and exclude the problem if the remaining chain equals that of a k-task. A problem that cannot be placed with confidence is excluded. The third is a manual adversarial review of 273 problems that an earlier and coarser rule had excluded by algorithm family; it re-admitted 191 and kept 82 out.

Three independently written checkers confirm on the final file that no excluded problem is present, that no generated row has an exam chain, and that no JAXBench task name appears. As a text-level check, the training reference closest to any JAXBench reference has a line similarity of 0.31, whereas JAXBench references resemble each other with a median of 0.66. We stress that this rule excludes identical computations, not similar ones. Section V quantifies how close the remaining training rows are to the tasks the adapter passes.

D. Training data

A training row is a conversation rendered by the evaluation harness's own prompt function, so the model sees byte-identical formatting in training and at test time. The target is always a kernel that the oracle judged correct on a TPU. Teacher reasoning is never a target. A repair row contains the problem, one or two failed kernels with the oracle's verbatim feedback, and the kernel that finally passed; the loss is computed on the final kernel only, so failed kernels serve as context.

Teacher. Kimi K3.1 [12], an open-weights model, wrote kernels for 1,672 problems from a fixed prompt containing kernel rules and about 25k tokens of Pallas documentation. On first attempts judged on the TPU, 434 passed and 267 were correct but slow. A second round, prompted with the earlier kernels and the oracle's feedback, added 235 passing problems, and a third round on 486 problems that had failed twice added 103. No closed-model output is used as a training target.

Rows at the benchmark's scale. An adapter trained on teacher rows alone failed the benchmark's matmul tasks in a consistent way: it loaded the whole weight into VMEM. The cause was a scale mismatch. The teacher's matmul problems had a median weight of 2 MB, for which loading the weight whole is correct, while JAXBench's matmul tasks use weights of 268 MB, for which it overflows. We therefore added rows at production scale in three ways. First, we enlarged 198 matmul-dominant problems until the weight was at least 64 MB and re-ran their existing kernels; 50 still passed. Second, the teacher repaired the others using the oracle's feedback and a tiling policy that we measured on the TPU (Fig. 3): a VMEM limit of 124 MiB and tiles of $(2048, 2048, 1024)$ along the output rows, output columns and contraction axis. Third, a template generator emits matmul, batched-matmul and row-reduction kernels that follow the same policy with varied elementwise epilogues, input layouts and sizes. Of 453 generated candidates judged at full size on the TPU, 434 passed. Nearly every generated row appears a second time as a surface variant with renamed arguments, configuration keys and docstring, admitted only when its syntax tree equals the judged kernel's after renaming; a stratified sample of 91 variants passed the oracle 91 of 91.

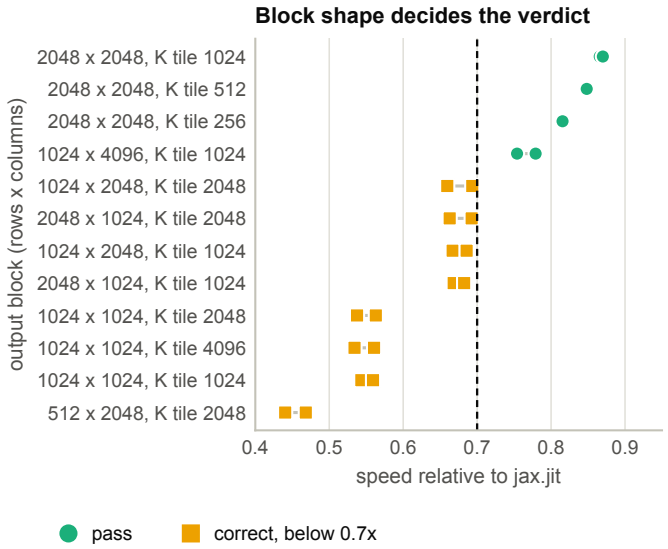


Fig. 3. The TPU sweep behind the tiling policy, for a matmul with 2,048 output rows. Only large output blocks clear the 0.7× bar; the same kernel with smaller blocks is correct but too slow.

Composition. Table I gives the final file. The teacher wrote 1,595 targets and the generator 709 (30.8%). There are 1,720 single-turn rows and 584 repair chains. For 348 rows (15%), all on non-matmul teacher problems, the target is correct but below the speed bar; slow matmul targets were removed because they demonstrate the untiled design that the production-scale rows are meant to replace. The longest row is 11,854 tokens.

E. Training and serving

We train a LoRA adapter of rank 64 and scale 128 on the attention and MLP projections of Gemma 4 E4B (139.5M

TABLE I
COMPOSITION OF THE TRAINING SET.

Part	Rows
Teacher base set, after contamination filtering	1,172
Teacher kernels at production scale	217
Teacher third-round kernels (repair and single-turn)	206
Template rows (originals and surface variants)	709
Total	2,304

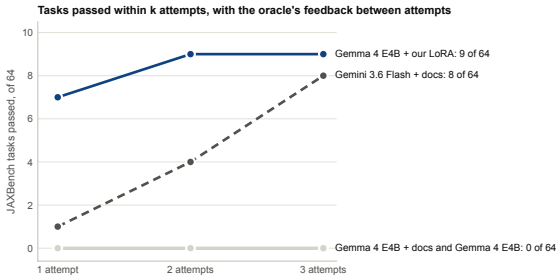


Fig. 4. Tasks passed within one, two and three attempts, over all 64 tasks. On the 58-task primary set Flash passes 7; its eighth pass is on a task outside that set.

trainable parameters), for 2 epochs at a peak learning rate of 10^{-4} with AdamW, 10% warm-up, cosine decay and gradient accumulation of 4, which gives 1,152 optimizer steps. Training took 89 minutes on one A100 80 GB and cost under 6 US dollars. The mean training loss was 0.157 in the first epoch and 0.075 in the second. The scored checkpoint, the end of epoch 2, was fixed before training began.

The baselines run the official 4-bit quantization-aware GGUF build of the model under Ollama, which cannot load a LoRA adapter at run time, so the adapter has to be folded into the weights. In our first attempts we trained on the bf16 weights and re-quantized the sum to 4 bits. This erased most of the adapter, because typical LoRA updates are far smaller than one 4-bit rounding step. Two changes fix it. We train against the dequantized 4-bit weights, so that the adapter is fitted to exactly the base it will be served with, and we store the 258 adapted tensors at 8 bits as the dequantized base plus the update, leaving every other tensor byte-identical to the official file. The trainer's and the server's token counts agree on all 64 first-turn prompts, and every evaluation row records the digest of the served model.

F. Evaluation protocol

Tasks. We use JAXBench's 50 level-1 tasks and 14 level-2 tasks for which we wrote plain JAX references. The primary set is the 58 tasks whose reference we could verify on the TPU: all 50 level-1 tasks and 8 level-2 tasks whose reference matched Google's kernel. It contains 33 k-tasks and 25 p-tasks. The task set, the oracle and the metric were fixed before the first adapter was evaluated.

Budget. Every arm gets one first attempt and up to two repair attempts per task. A repair prompt contains the earlier attempts and the oracle's feedback line. A task stops at its first pass.

Arms. (i) Base Gemma 4 E4B with the reference only (*bare*). (ii) The same model with about 25k tokens of Pallas documentation prepended (*docs*). (iii) Gemma 4 E4B with our adapter, with the bare prompt. (iv) Gemini Flash (served as `gemini-3.6-flash`) with the docs prompt. The three Gemma arms decode greedily with thinking off and at most 8,192 output tokens, under the same Ollama version on the same GPU type. Flash runs at its default temperature with thinking on (a median of 5,518 thinking tokens per attempt) and up to 65,536 output tokens, because it loops at temperature zero. Flash is therefore a reference point with a different model, prompt and decoding regime, and not a controlled comparison.

Metrics and tests. The primary metric is *pass within 3*: the number of tasks with a `pass` on any attempt. Secondary metrics are *correct within 3* (a `pass` or `correct_slow` on any attempt), pass at the first attempt, and parity (a correct kernel at 0.97× XLA or faster). Arms are compared on paired per-task outcomes with an exact two-sided McNemar test [14], and we report Holm-adjusted values [15] over the three comparisons we make. Proportions are given with 95% Wilson intervals [16].

IV. RESULTS

A. Main result

Table II and Fig. 4 give the result. Base Gemma passes no task. It also produces no correct kernel at any speed: with the bare prompt every one of its attempts fails the kernel check or compilation, and adding documentation changes only which early stage rejects them. The adapter, given the same bare prompt, passes 9 of 58 tasks. All nine discordant tasks favor the adapter, so the improvement over the base model is significant ($p = 0.004$; Holm-adjusted $p = 0.012$, Table III). The result is the same whether the base model has documentation or not.

Against Gemini Flash with documentation, the adapter passes 9 tasks to Flash's 7. The two sets are almost disjoint, with only RMSNorm (12p) in common, and the 8-to-6 split of discordant tasks is far from significant ($p = 0.79$). On this benchmark and budget the two systems are statistically tied on the pass metric. The tie should not be read as equal ability, and the next subsection shows why.

On first attempts the adapter passes 7 tasks and Flash 1. The unadjusted test gives $p = 0.031$, and $p = 0.06$ after Holm adjustment, so we report this as suggestive only. The adapter rarely benefits from repair: its count rises from 7 to 9 at the second attempt and not at all at the third, while Flash rises from 1 to 3 to 7.

B. Correct versus fast

The pass metric requires a kernel to be both correct and fast, and the two models fail different halves of that requirement (Figs. 5 and 6). Flash writes a numerically correct kernel for

TABLE II

JAXBENCH RESULTS ON THE 58-TASK PRIMARY SET, THREE ATTEMPTS PER TASK. A PASS IS A CORRECT KERNEL AT 0.7× XLA OR FASTER. BRACKETS GIVE 95% WILSON INTERVALS FOR THE PASS RATE.

Arm	Pass within 3	95% interval	Correct within 3	Pass at 1	Parity	k-tasks (33)	p-tasks (25)
Gemma 4 E4B, bare	0	[0.0%, 6.2%]	0	0	0	0	0
Gemma 4 E4B + docs	0	[0.0%, 6.2%]	0	0	0	0	0
Gemini Flash + docs	7	[6.0%, 22.9%]	36	1	4	5	2
Gemma 4 E4B + our LoRA	9	[8.4%, 26.9%]	9	7	9	8	1

TABLE III

PAIRED COMPARISONS OF THE ADAPTER WITH THE OTHER ARMS ON THE 58 PRIMARY TASKS (EXACT TWO-SIDED MCNEMAR TEST).

Versus	Metric	Ours only	Other only	p
Base (bare or docs)	pass within 3	9	0	0.004
Flash + docs	pass within 3	8	6	0.79
Flash + docs	pass at 1	6	0	0.031
Flash + docs	correct within 3	1	28	$< 10^{-6}$

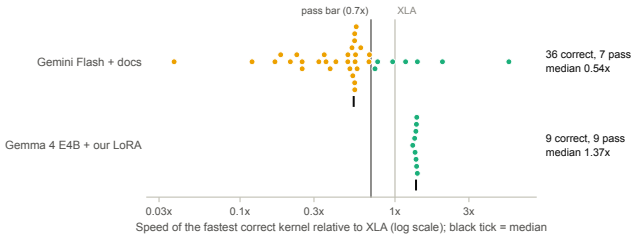


Fig. 5. Speed of the fastest correct kernel per task, for every task on which each model was correct within three attempts.

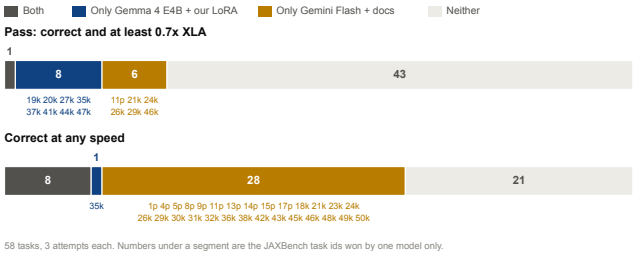


Fig. 6. Paired outcomes of the adapter and Gemini Flash with documentation on the 58 primary tasks, for the pass metric (top) and for correctness at any speed (bottom).

36 of the 58 tasks, four times as many as the adapter's 9, and on correctness alone it is far ahead (1 task correct only for the adapter, 28 only for Flash, $p < 10^{-6}$). But most of Flash's correct kernels are slow. The fastest correct kernel it produces per task has a median speed of 0.54× XLA, and only 7 of the 36 reach the bar.

The adapter shows the opposite profile. It is correct on only 9 tasks, but every correct kernel it writes passes, and all nine run faster than XLA, between 1.30× and 1.39×. It has learned one design that is fast when it applies, and it has not learned to fall back to a slow correct kernel when that design does not apply. Four of its passes (19k, 20k, 37k, 41k) satisfy the oracle's second correctness rule rather than the fixed tolerance; their error equals that of XLA's own compiled reference.

C. What the adapter writes

Eight of the nine passes are k-tasks led by a matrix multiply followed by elementwise operations; the ninth is RMSNorm. The eight matmul kernels share one structure, shown in Fig. 7 for task 41k. The grid iterates over tiles of the output rows, output columns and contraction axis. A float32 accumulator in VMEM is cleared on the first contraction step, each step adds one partial product, and the last step applies the bias and activation and writes the output tile. The kernel also raises the VMEM limit to 124 MiB. This is the tiling policy of the production-scale training rows, applied to chains of operations that are not in the training set.

D. Where the adapter fails

Fig. 8 shows how far each arm gets through the oracle. The adapter's failures fall into four groups.

Compilation. Of the 58 first attempts, 35 fail to compile, and repair does not help: across all 64 tasks, in 80 of the 81 cases where a compile failure was followed by another attempt, the next attempt also failed to compile. The oracle's one-line feedback is enough to correct a memory layout but not a misuse of the API.

Row reductions. Eight first attempts overflow VMEM. On repair two become passes, and six fit in memory but compute wrong values. All six are a matmul followed by an operation over a whole output row (log-sum-exp, softmax, a row sum or a row maximum), where the epilogue must see a complete row while the contraction axis is tiled.

Convolutions. None of the 24 attempts on the 8 convolution-led k-tasks gets past compilation.

Production kernels. Among the 25 p-tasks only RMSNorm passes. The others are attention variants, mixture-of-experts routing and state-space kernels, and almost all of the adapter's attempts on them fail to compile.

V. DISCUSSION AND LIMITATIONS

The headline result supports a narrow claim. Fine-tuning on a few thousand verified examples moved a 4B model from never producing a working Pallas kernel to producing fast ones for one family of tasks. The following points bound that claim, and we consider them part of the result.

The training data was informed by the benchmark. No JAXBench task, reference or kernel is in the training set, and

```
def _kernel(x_ref, weight_ref, bias_ref, out_ref, acc_ref):
    k = pl.program_id(2)

    @pl.when(k == 0)
    def _zero_accumulator():
        acc_ref[...] = jnp.zeros_like(acc_ref)

    acc_ref[...] += jnp.dot(x_ref[...], weight_ref[...], preferred_element_type=jnp.float32)

    @pl.when(k == pl.num_programs(2) - 1)
    def _store_result():
        value = acc_ref[...] + bias_ref[...]
        value = jax.nn.relu(value)
        out_ref[...] = value.astype(out_ref.dtype)

def workload(x, weight, bias):
    m, k = x.shape
    n = weight.shape[1]
    tm = _tile(m, 2048, 16)
    tn = _tile(n, 2048, 128)
    tk = _tile(k, 1024, 128)
    return pl.pallas_call(
        _kernel,
        out_shape=jax.ShapeDtypeStruct((m, n), x.dtype),
        grid_spec=pltpu.PrefetchScalarGridSpec(
            num_scalar_prefetch=0,
            grid=(m // tm, n // tn, k // tk),
            in_specs=[
                pl.BlockSpec((tm, tk), lambda i, j, kk: (i, kk)),
                pl.BlockSpec((tk, tn), lambda i, j, kk: (kk, j)),
                pl.BlockSpec((tn,), lambda i, j, kk: (j,)),
            ],
            out_specs=pl.BlockSpec((tm, tn), lambda i, j, kk: (i, j)),
            scratch_shapes=[pltpu.VMEM((tm, tn), jnp.float32)],
        ),
        compiler_params=pltpu.CompilerParams(
            dimension_semantics=("parallel", "parallel", "arbitrary"),
            vmem_limit_bytes=124 * 1024 * 1024,
        ),
    )(x, weight, bias)
```

Fig. 7. The adapter's first attempt on task 41k (a 4096×8192 input times an 8192×8192 weight, plus bias, then ReLU), which passes at $1.37\times$ XLA. Imports and the helper `_tile`, which picks the largest aligned tile that divides an axis, are omitted, and the `jnp.dot` call is joined onto one line.

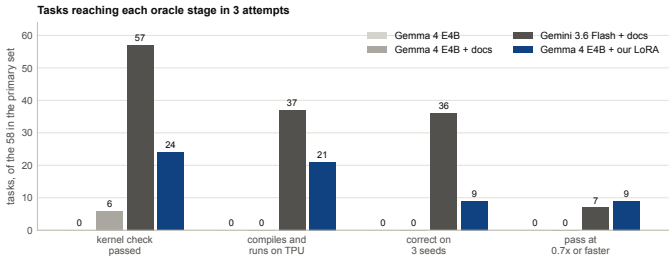


Fig. 8. Number of primary-set tasks that reach each oracle stage within three attempts.

problems with the same computation chain as a k-task were removed. However, the adapter reported here is the last of several that we evaluated on JAXBench during development, and what we learned from earlier evaluations shaped the data. The production-scale rows exist because an earlier adapter failed the matmul tasks, they were sized to the benchmark's matmul scale, and 16 generated rows have exactly the input shapes of matmul tasks the adapter passes. The passed matmul tasks differ from their nearest generated training kernel by one elementwise operation in most cases (two for 47k, four for 27k). The correct reading of the eight matmul passes is therefore that the model applies a learned tiling design to new epilogues at familiar shapes, and not that it designs kernels for unfamiliar problems.

Sensitivity to near neighbours. The contamination rule was relaxed from family level to instance level after results of earlier adapters had been seen, which re-admitted 191 problems. If the 19 primary tasks that have a re-admitted or same-family training neighbour are set aside, 39 tasks remain. On these the adapter passes 5, the base model 0 (exact McNemar $p = 0.0625$) and Flash 7 (4 tasks only ours, 6 only Flash, $p = 0.75$). The direction of the effect is unchanged, but with five discordant tasks the comparison with the base model no longer reaches the 0.05 level.

Partial tiles. Every template row has power-of-two sizes, so its tiles always divide the arrays, and the adapter's kernels assume this. We re-ran the eight passed matmul tasks at two shapes absent from training, $(2048, 7168, 9216)$ and $(3072, 7680, 8704)$. The adapter passed 3 of 8 at the first and 0 of 8 at the second. Its kernels either compute `size // tile` grid steps and never process the remainder, or choose block sizes that the TPU rejects. The skill is presently limited to sizes the learned tiles divide.

The Flash reference is imperfect. Flash differs from the adapter in model size, prompt, sampling and thinking budget,

and its control is incomplete on two tasks that only the adapter passes. On 35k its first attempt halted a TPU core; re-judged alone it is correct at 0.22× XLA, and its two repair attempts were never generated. On 19k it received two attempts instead of three. Excluding both tasks, each model passes 7 of 56 with a 6-to-6 split of discordant tasks ($p = 1$), so the conclusion of a tie does not depend on them.

Testing is not verification. Reading the training targets after training, we found 5 rows (0.2%) whose kernels pass the oracle through shortcuts that depend on the input distribution, such as a mean substituted for a median, or a polynomial that is accurate only on the reference's input range. Three seeds from a fixed distribution cannot rule these out. The adapter was trained on them; the published dataset excludes them.

Small numbers and a single sample. The primary set has 58 tasks, each arm contributes one chain per task, and Gemma decodes greedily. The Wilson intervals in Table II are wide and overlap for the adapter and Flash. Three attempts per task is also far below the 144 used in the JAXBench paper, so none of our numbers is comparable to its reported solve rates.

Mixed signals in the data. About 31% of rows come from one generator, 15% of targets are below the speed bar, and most teacher kernels use the default VMEM limit while generated rows set 124 MiB. The held-out development loss (0.0002) is on template items and measures template recall, not kernel-writing ability.

VI. Conclusion

A 4B open model that could not write a single working Pallas kernel, with or without documentation, learned from 2,304 oracle-verified examples to write tiled matmul kernels that run 1.3× to 1.4× faster than XLA. On JAXBench's 58 primary tasks with three attempts it passes 9, a significant improvement over the base model's 0 and a statistical tie with Gemini Flash given documentation. The tie hides a clear difference in kind: Flash is correct four times as often and usually slow, and the adapter is rarely correct but fast whenever it is. The adapter's competence is confined to the matmul family at sizes its tiles divide, most of its failures never compile, and the training data was shaped by the benchmark it is scored on. The next steps follow directly from the failures: training rows with partial tiles, rows that teach whole-row reductions under a tiled contraction, and feedback that explains compile errors rather than reporting them.

Reproducibility

Code, data and results are in the repository `Art4v/gemma-hackathon-research`. The oracle is in `oracle/`. The release `dataset-final` contains the training file as trained (2,304 rows), the published file without the five shortcut rows (2,299 rows), the development set, a datasheet and checksums. The raw generations and verdicts of every arm are in `paper/data/runs/`, and `paper/analysis/jaxbench.py` recomputes every number in Tables II and III from them using only the Python standard library.

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